

In Hot Water: Clarifying British Prices in the Wake of Brexit

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Non-trade policy barriers remain poorly understood

- ▶ International goods markets seem on a **path of fragmentation**.
 - ▶ **Shallow integration:** goods markets disrupted by trade policy barriers
 - ▶ e.g. import tariffs and quotas in 2018, 2025, ...
 - ▶ **Deep integration:** trade-facilitating markets disrupted by non-trade-policy barriers
 - ▶ e.g. regulation of transport markets, use of financial sanctions.
 - ▶ Assessing the **importance of non-trade policy barriers** is **empirically hard**:
 - ▶ **Relevant measure** when barriers relate to efficacy of trade-facilitating markets?
 - ▶ **Seperate change in barriers** from endogenous response of agents?
 - ▶ **Mechanism** behind the change in the barriers?
 - ▶ Consider the EU-UK **Trade and Cooperation Agreement (TCA)**:
 - ▶ ensures tariff-free trade between the EU and UK...
 - ▶ ... but EU → UK trade was 25% lower in 2022 (Freeman et al., 2024).
- ↪ With **upcoming evaluation of TCA**, a better understanding non-trade-policy barriers is needed.

This paper aims to provide some clarity using... bottled water

This paper aims to:

1. estimate increase in **EU** → **UK non-trade policy barriers** following the introduction of the TCA
2. quantify the **contribution** of one mechanism, i.e., an increase in freight rates for road transport.

How do we answer **abovementioned concerns**?

- ▶ **Relevant measure** when barriers relate to efficacy of trade-facilitating markets?
 - ↳ Compare change in **marginal cost** of imported products in UK to the marginal cost in BE and FR.
- ▶ **Seperate change in barriers** from endogenous response of agents?
 - ↳ We focus on the **bottled water industry**.
 - ↳ (1) Production relocation is restricted, (2) Short value chain
- ▶ **Mechanism** behind the change in the barriers?
 - ↳ We focus on the change in **freight rates for road transportation**.
 - ↳ (1) Customs checks and regulatory change (2) Important mode of transportation

What do we do and find?

I will discuss **three results**:

1. Response of **UK consumer prices** after the introduction of the TCA:

- ▶ Prices of **imported** products $\uparrow \approx 17\%$,
- ▶ Prices of **domestic** products $\uparrow \approx 15\text{-}20\%$.

2. Response of **freight rates** for road transportation:

- ▶ **International** routes $\uparrow \approx \text{€}850$,
- ▶ **domestic** routes $\uparrow \approx \text{€}150$.

3. What is the **overall increase** in **non-trade-policy barriers** and what is the **contribution of** changes in freight rates?

- ▶ We estimate a model of demand and supply of the bottled water industry
- ↪ Non-trade policy barriers **increased by 17%** following the TCA (preliminary!)
- ↪ Transport costs **account for ~ 25%** of this increase (preliminary!)

We contribute to three literatures

- ▶ **Deep trade agreements and non-trade policy barriers:** Kee, Nicita, and Olarreaga (2009), Chen and Novy (2011), Baier, Bergstrand, and Feng (2014), Fontagné et al. (2015), and Dhingra, Freeman, and Huang (2023)
 - ▶ uses trade flows as the measure $\implies$ little guidance on which barriers matter most.
 - ▶ This paper uses prices as measure and quantifies contribution of changes in freight rates
- ▶ **Endogenous transport costs:** Hummels, Lugovskyy, and Skiba (2009), Behrens and Picard (2011), Pascali (2017), Ishikawa and Tarui (2018), Brancaccio, Kalouptsi, and Papageorgiou (2020) and Wong (2022)
 - ▶ considers the joint determination of trade flows and freight rates in equilibrium
 - ▶ This paper shows that freight rates endogenously responded following the TCA.
- ▶ **Effects of Brexit:** (Steinberg, 2019; Graziano, Handley, and Limão, 2021; Fernandes and Winters, 2021; Breinlich et al., 2022; Bakker et al., 2022; Corsetti, Crowley, and Han, 2022; Ahmad et al., 2023; Freeman et al., 2024)
 - ▶ focuses on uncertainty and news effects before the TCA
 - ▶ This paper focuses on the effects after the TCA and quantifies the contribution of one channel.

DATA AND SETTING

Data sources

We combine **three data sources** and some auxiliary data:

1. **Household-level scanner data** on non-alcoholic beverages from GfK and Kantar
 - ▶ **Spatial:** BEL, FRA, and UK
 - ▶ **Time:** Daily from 2010 until 2022, aggregated to monthly level
 - ▶ **Water variety:** A combination of brand-flavored-package-bottle size
2. Based on Directive 2009/54/EC, we hand-collect **production locations** of bottled water
3. Route-level **freight rates** from Upply
 - ▶ **Spatial:** From ≈ 20 water sources to ≈ 40 major population centers
 - ▶ **Time:** Monthly from 2018 until 2023
 - ▶ **Type:** Spot price for fully loaded trailer
4. **Auxiliary data** on wages in retail sector and intermediate input prices of bottle packages.

Data - Consumption data

Table 1: Sample overview

Variable	Overall	BEL	FRA	UK
Nr. Households				
· NARTD	53,122	5,303	17,915	29,904
· Water	40,219	4,662	16,326	19,231
Transactions ('1,000)	7,319	1,014	3,293	3,012
Nr. Products	660	297	265	214
Average consumer price				
· (LCU)	-	0.37	0.28	0.33
· (EUR)	-	0.37	0.28	0.40
Inside good share (rel. NARTD)				
· Expenditure	-	0.22	0.30	0.13
Nr. Production locations				
· All	122	71	67	48
· Local	-	10	56	35
Origin (volume share)				
· Belgium	-	0.211	0.001	0.000
· France	-	0.473	0.902	0.075
· UK	-	0.000	0.000	0.743
· Other	-	0.103	0.066	0.021
· Unknown	0.091	0.213	0.031	0.161

Data - Production locations

Figure 1: Production locations

(a) All sources



(b) Sources selling to UK



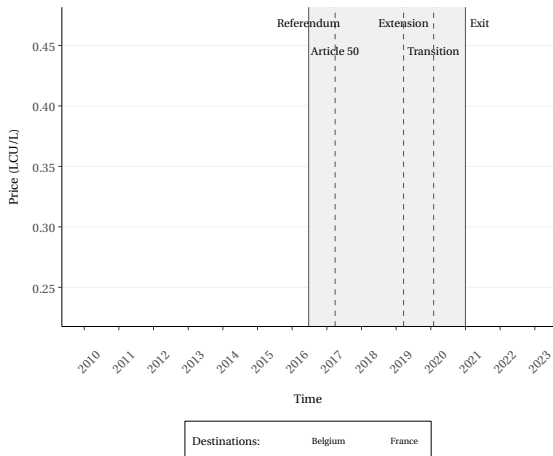
Data - Consumption data

Table 2: Sample overview

Variable	Overall	BEL	FRA	UK
Nr. Households				
· NARTD	53,122	5,303	17,915	29,904
· Water	40,219	4,662	16,326	19,231
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REDUCED-FORM EVIDENCE

The timeline of Brexit



Brexit was a **lengthy process**:

1. UK decides to leave the EU via **referendum of June 23rd, 2016 ...**
2. followed by a **painfully slow process**:
 - ▶ **Article 50** was triggered in 2017
 - ▶ Missed deadlines and **extensions** in 2019
 - ▶ **Transition period** in January 2020 towards exit
3. The **Trade and Cooperation Agreement** governs EU-UK trading relations from January, 2021 onwards:
 - ▶ **FTA** subject to rules-of-origin rules
 - ↳ But: **Regulatory changes** and customs checks.

Finding 1: Estimating the post-TCA response of consumer prices

The **response of UK consumer prices** of **imported** and **domestically produced** bottled water:

1. UK consumer prices of **imported** water:

$$\ln(p_{j,dt}^{\text{LCU}}) = \sum_t \beta_t \left(\mathbb{1}(d = \text{UK}) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{j,d} + \lambda_{j,t} + \varepsilon_{j,dt}$$

where

- ▶ $\theta_{j,d}$ are product-destination fixed effects.
- ▶ $\lambda_{j,t}$ are product-time fixed effects.
- ▶ Sample: $(j, d) = \{\text{Evian-BEL, Evian-FRA, Evian-UK, Buxton-UK}\}$

2. UK consumer prices of **domestic produced** water:

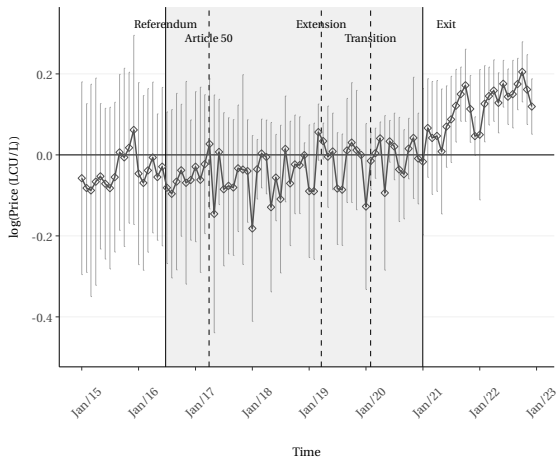
$$\ln(p_{j,t}^{\text{LCU}}) = \sum_t \beta_t \left(\mathbb{1}(j = \text{UK}) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_j + \lambda_t + \varepsilon_{j,t}$$

where

- ▶ θ_j are product fixed effects (only one destination).
- ▶ λ_t are time fixed effects.
- ▶ Sample: $(j) = \{\text{Evian-BE, Evian-FR, Evian-UK, Buxton-UK}\}$

Finding 1: UK consumer prices of imported water $\uparrow \approx 17\%$

Figure 2: Treatment effect



► Specification:

$$\ln(p_{j,dt}^{\text{LCU}}) = \sum_t \beta_t \left(\mathbb{1}(d = \text{UK}) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{j,d} + \lambda_{j,t} + \varepsilon_{j,dt}$$

► Sample: {Evian-BE, Evian-FRA, Evian-UK, Buxton-UK}

► Findings:

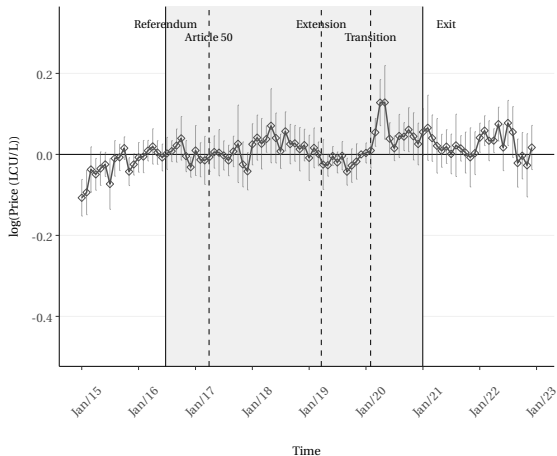
- Prior to the TCA, UK prices track BE and FR prices closely.
- Two years after the TCA, UK prices are 17% higher.

► GBP/EUR

► Covid infections

Finding 1: Consumer prices in BEL $\approx$ Consumer prices in FRA

Figure 3: Placebo effect



► Specification

$$\ln(p_{j,dt}^{\text{LCU}}) = \sum_t \beta_t \left(\mathbb{1}(d = \text{UK}) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{j,d} + \lambda_{j,t} + \varepsilon_{j,dt}$$

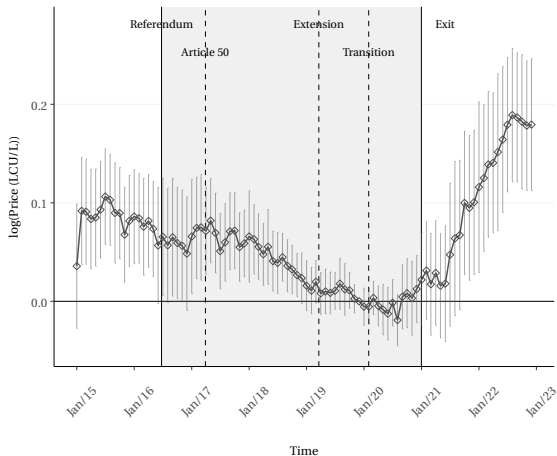
► **Sample:** {Evian-BE, Evian-FR, ~~Evian-UK~~,
~~Buxton-UK~~},

► Findings:

- **Prior to the TCA**, BE prices track FR prices closely.
- BE have not diverged from FR prices **after the TCA**,

Finding 1: UK consumer prices of domestically produced water $\uparrow$ $\approx 15 - 20\%$

Figure 4: Treatment effect



► Specification

$$\ln(p_{j,t}^{\text{LCU}}) = \sum_t \beta_t \left(\mathbb{1}(d = \text{UK}) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_j + \lambda_t + \varepsilon_{j,t}$$

► **Sample:** {~~Evian-BE~~, Evian-FR, ~~Evian-UK~~, Buxton-UK}

► Findings:

- **Prior to the TCA**, UK prices of domestically produced water on a downward trend.
- Two years **after the TCA**, UK prices of domestically produced are 15-20% higher.

Finding 2: Estimating the post-TCA response of freight rates

We consider the response of **international** and **domestic** UK-bound routes:

1. Freight rates of **international** UK-bound routes:

$$t_{o,d,t} = \sum_t \beta_t \left(\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{o,d} + \lambda_{o,t} + \varepsilon_{o,d,t}$$

where

- ▶ $\theta_{o,d}$ are origin-destination fixed effects.
- ▶ $\lambda_{o,t}$ are origin-time fixed effects.
- ▶ Sample: $(o, d) = \{\text{Evian-Brussels, Evian-Paris, Evian-Cambridge, Buxton-Cambridge}\}$

2. Freight rates of **domestic UK-bound routes**:

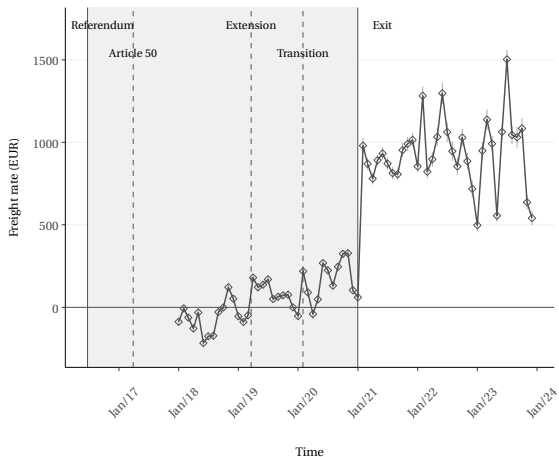
$$t_{o,d,t} = \sum_t \beta_t \left(\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{o,d} + \lambda_t + \varepsilon_{o,d,t}$$

where

- ▶ $\theta_{o,d}$ are origin fixed effects.
- ▶ λ_t are time fixed effects.
- ▶ Sample: $(o, d) = \{\text{Evian-Brussels, Evian-Paris, Evian-Cambridge, Buxton-Cambridge}\}$

Finding 2: International UK-bound freight rates $\uparrow \approx 850\text{EUR}$

Figure 5: Treatment effect



► Specification:

$$t_{o,dt} = \sum_t \beta_t \left(\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{o,d} + \lambda_{o,t} + \varepsilon_{o,dt}$$

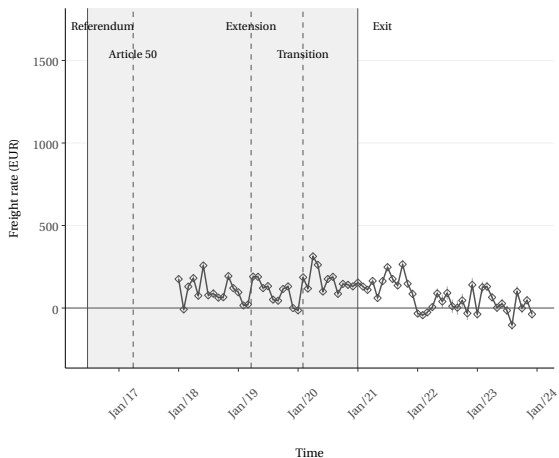
► Sample: {Evian-Brussels, Evian-Paris, Evian-Cambridge, ~~Buxton-Cambridge~~}

► Findings:

- **Prior to the TCA**, rates on international UK-bound rates are a mild upward trend.
- Two years **after the TCA**, rates on international UK-bound rates are €850 more expensive.

Finding 2: Freight rates on BEL-bound $\approx$ freight rates on FRA-bound routes

Figure 6: Placebo effect



► **Specification:**

$$t_{o,d,t} = \sum_t \beta_t \left(\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{o,d} + \lambda_{o,t} + \varepsilon_{o,d,t}$$

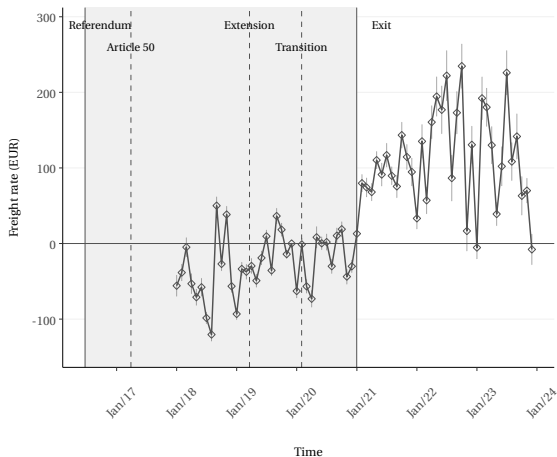
► **Sample:** {Evian-Brussels, Evian-Paris, ~~Evian-Cambridge~~, ~~Buxton-Cambridge~~}

► **Findings:**

- **Prior to the TCA**, rates on BE- and FR-bound routes closely track each other
- Rates on BE- and FR-bound routes closely track each other **after the TCA**.

Finding 2: Domestic UK-bound freight rates $\uparrow \approx 100 - 150\text{EUR}$

Figure 7: Placebo effect



► **Specification:**

$$t_{o,dt} = \sum_t \beta_t \left(\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12) \right) + \theta_{o,d} + \lambda_t + \varepsilon_{o,dt}$$

► **Sample:** { ~~Evian-Brussels~~, Evian-Paris, ~~Evian-Cambridge~~, Buxton-Cambridge }

► **Findings:**

- **Prior to the TCA**, rates on domestic UK-bound routes evolve like those on domestic BE/FR-bound routes.
- **After the TCA**, rates on domestic UK-bound routes are $\approx \text{€ } 150$ more expensive.

A STRUCTURAL MODEL

Purpose of the model

We leverage a **structural model** for two reasons:

1. to map changes in consumer prices into **changes in non-trade policy barriers**
 - ▶ separating changes in **destination-specific markups** from changes in destination-specific marginal costs
 - ▶ controlling for changes in **other destination-specific non-traded costs**, e.g. distribution costs.
2. to quantify **the contribution of changes in freight rates** to changes in non-trade policy barriers
 - ▶ by estimating the **cost-function** of supplying bottled water
 - ▶ by calculating a **counterfactual scenario** where UK-bound freight rates track BEL- and FRA-bound rates.

Sketch of the (simple) model

The structural model has the following **building blocks**:

- ▶ **Preferences**: Consumers make a static discrete choice about which water to consume.
- ▶ **Market structure**: Consumer prices are determined via:
 - ▶ **Upstream manufacturers** compete as oligopolists and set upstream wholesale prices
 - ▶ **Downstream retailers** perfectly pass on wholesale prices and add local distribution cost
- ▶ **Technology**: Production of bottled water is Leontief in:
 - ▶ Local distribution costs
 - ▶ Transportation costs to the destination market
 - ▶ Production costs which can have arbitrary returns to scale and scope

Preferences - Assumptions and specification

Consumers $i = 1, \dots, N_{dt}$ chooses among $j = 1, \dots, \mathcal{J}_{dt} + 1$ products by solving:

$$\max_{j \in \mathcal{J}_{dt} + 1} U_{ij,dt} = \alpha \ln(P_{j,dt}) + \beta' X_{j,dt} + \xi_{j,dt} + \zeta_{g(j)} + \tilde{\varepsilon}_{ij,dt}$$

where

- ▶ $\alpha \ln(P_{j,dt})$ captures price sensitivity
- ▶ $\beta' X_{j,dt} = \{\theta_{j,d}, \theta_{dt}\}$ are absorb product characteristics.
- ▶ $\xi_{j,dt}$ is a product-level demand shifter.
- ▶ $\zeta_{g(j)}$ is a random effect with $\mathcal{G} = \{\text{branded, private label}\}$.
- ▶ $\tilde{\varepsilon}_{ij,dt}$ captures taste for products by consumers. We assume:

$$\zeta_{g(j)} + \tilde{\varepsilon}_{ij,dt} \sim F(\varepsilon) = \exp \left\{ - \sum_{g \in \mathcal{G}} \left(\sum_{j \in \mathcal{J}_{g,dt}} \exp[-\varepsilon / (1 - \sigma)] \right) \right\}^{1-\sigma}$$

Preferences - Results

Table 3: Demand estimates

	OLS	IV
$\ln(s_{j,dt}/s_{0,dt})$	(1)	(2)
$\log(p_{j,dt})$	-0.0287* (0.016)	-1.69** (0.772)
$\log(s_{jlg,dt})$	1*** (0.003)	0.86*** (0.143)
N	28,275	28,275
1 st Stage F-stat : $\log(p_{j,dt})$	-	41.20
1 st Stage F-stat : $\log(s_{jlg,dt})$	-	25.95
$\theta_{j,d}$	✓	✓
λ_{dt}	✓	✓
Median Own Price Elasticity	-	-12.03
Median Market Price Elasticity	-	-0.35

Notes: Standard errors clustered at the product-destination level.
Significance at the 0.1*, 0.05**, 0.01*** levels.

- ▶ With assumption on $\zeta_{g(j)} + \tilde{\varepsilon}_{ij,dt}$, **estimable equation** is:

$$\ln\left(\frac{s_{j,dt}}{s_{0,dt}}\right) = \alpha \ln(P_{j,dt}) + \boldsymbol{\beta}' \mathbf{X}_{j,dt} + \sigma \log(s_{jlg,dt}) + \xi_{j,dt}$$

- ▶ The **results** are:
 - ▶ **OLS** yields inelastic demand
 - ▶ We employ the following **instruments**:
 - ▶ $P_{j,dt}$: international hausman instrument (F-stat: 41.20)
 - ▶ $s_{jlg,dt}$: number of products per nest (F-stat: 25.95)
 - ▶ **IV** yields very elastic own-price elasticities at ≈ -12 .

Estimating the cost function - Assumptions and specification

With **upstream Bertrand-Nash** and **perfect pass-through downstream retailers**, we have that:

- Using the FOCs, we **separate markups from marginal costs**:

$$\underbrace{c_{dt}^r + c_{dt}^m}_{\text{Marginal cost}} = \underbrace{p_{dt}^r}_{\text{Retail price}} + \underbrace{(\Delta_{dt} \odot \Omega_{dt}^w)^{-1} \cdot s_{dt}(p^r; \Theta^d)}_{\text{Manufacturer markup}}$$

- By parameterizing backed out marginal costs, we **decompose changes in marginal costs**:

$$\underbrace{c_{j,dt}^r + c_{j,dt}^m}_{\text{Marginal costs}} = \underbrace{\omega_{j,t}}_{\text{production costs}} + \underbrace{\beta \left(\frac{1}{k} \sum_k t_{s(j),dt-k} \right)}_{\text{Transport costs}} + \underbrace{\gamma w_{dt}^r}_{\text{Distribution cost}} + \eta_{j,dt}$$

- By feeding in **counterfactual** $t_{s(j),dt-k}^{\text{counter}}$, we can recompute $c_{j,dt}^r + c_{j,dt}^m$ and $p_{j,dt}$.

Estimating the cost function - Results

Table 4: Cost function estimates

$\hat{m}c_{j,dt}$	k = 0	k = 1
$\frac{1}{k} \sum_k t_{j,dt-k}$ (1'000 EUR)	-0.0122 (0.034)	0.124** (0.060)
N	28,853	28,853
Adj. R^2	0.71	0.72
$\omega_{j,t}$	✓	✓

Notes: Standard errors clustered at the product-destination level.
Significance at the 0.1*, 0.05**, 0.01*** levels.

- Estimate the following **equation**:

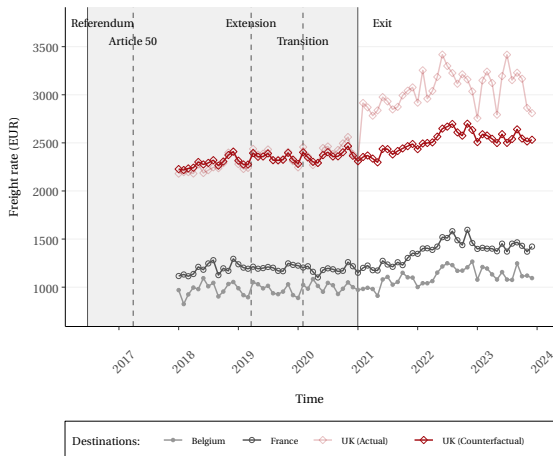
$$\widehat{c_{j,dt}^r + c_{j,dt}^m} = \omega_{j,t} + \beta \left(\frac{1}{k} \sum_k t_{s(j),dt-k} \right) + \eta_{j,dt}$$

- We obtain the following **results**:

- **Contemporaneously** uncorrelated
- When accounting for **lags**:

$$t_{s(j),dt-k} \uparrow \in 1000 \implies \widehat{c_{j,dt}^r + c_{j,dt}^m} \uparrow \in 0.12$$

Specifying the countefactual



- ▶ Estimate the following **regression**:

$$t_{o,dt} = \theta_{o,d} + \lambda_t + \varepsilon_{o,dt}$$

- ▶ Limit the **sample** to:

- ▶ Belgium and France: 2018-2023
- ▶ UK: 2018-2021

- ▶ Set **counterfactual** UK-bound freight rates as:

$$t_{o,dt}^c = \hat{\theta}_{o,d} + \hat{\lambda}_t$$

Clarifying British Prices in the Wake of Brexit - Imported products

y	$\ln(p_{jt})$	$\ln(\hat{m}c_{jt})$	$\ln(\hat{\mu}_{jt})$	$\ln(mc_{jt}^{\text{Counter}})$	$\ln(p_{jt}^{\text{Counter}})$
	(1)	(2)	(3)	(4)	(5)
$T_{i,d} \times \mathbb{1}(t = 2018)$	-0.0337** (0.0154)	-0.0339** (0.0154)	-0.0323** (0.0154)	-0.0191 (0.0132)	-0.0189 (0.0165)
$T_{i,d} \times \mathbb{1}(t = 2019)$	-	-	-	-	-
$T_{i,d} \times \mathbb{1}(t = 2020)$	0.00812 (0.0114)	0.00645 (0.0114)	0.0213* (0.0118)	0.00449 (0.0118)	0.00567 (0.0121)
$T_{i,d} \times \mathbb{1}(t = 2021)$	0.109*** (0.0203)	0.108*** (0.0192)	0.117*** (0.0329)	0.0672*** (0.0193)	0.0871*** (0.0194)
$T_{i,d} \times \mathbb{1}(t = 2022)$	0.171*** (0.0358)	0.17*** (0.0336)	0.182*** (0.0528)	0.116*** (0.0271)	0.123*** (0.0282)
Median of LHS	0.91	0.80	0.10	0.79	0.88
$\theta_{i,d}$	✓	✓	✓	✓	✓
$\lambda_{i,t}$	✓	✓	✓	✓	✓
R^2	0.377	0.378	0.361	0.193	0.236
No. obs	378	378	378	378	378

► Same **specification** as before:

$$\ln(y_{j,d,t}) = \sum_t \beta_t (\mathbb{1}(d = UK) \times \mathbb{1}(t \neq 2019M12)) + \theta_{j,d} + \lambda_{j,t} + \varepsilon_{j,d,t}$$

► **Actual** results:

$mc_{j,d,t}$ of imported products 14% ↑

► **Counterfactual** results:

► $mc_{j,d,t}$ of imported products 12% ↑

► $p_{j,d,t}$ of imported products 14.4% ↑

► Domestically produced

Notes: Standard errors clustered at the product-destination level.

Significance at the 0.1*, 0.05**, 0.01*** levels.

CONCLUDING REMARKS

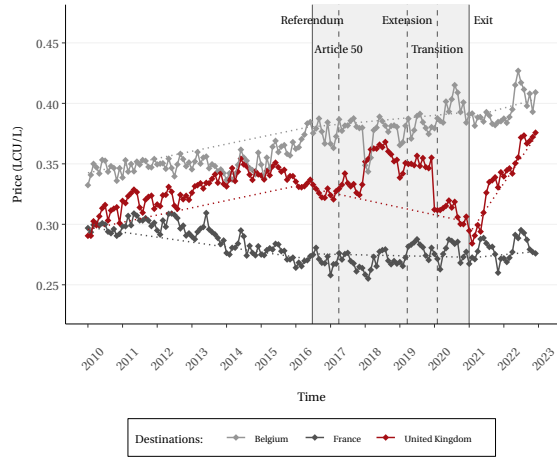
Concluding remarks

- ▶ Renewed **fragmentation** of international goods markets
 - ↳ non-trade policy barriers remain poorly understood (Head and Mayer, 2014).
- ▶ **This paper** leverages the bottled water industry
 1. to estimate the Brexit-induced increase in non-trade policy barriers
 2. quantify the contribution of one non-trade policy barrier, i.e. increased freight rates
- ▶ We have **three main findings**:
 - ▶ UK **consumer prices of imported water** increased by 17% relative to BEL and FRA prices.
 - ▶ **Freight rates on UK-bound routes** increased by 30% relative to BEL- and FRA-bound routes.
 - ▶ Increase in **freight rates explains** 25 – 30% of the increase in marginal costs of foreign products.

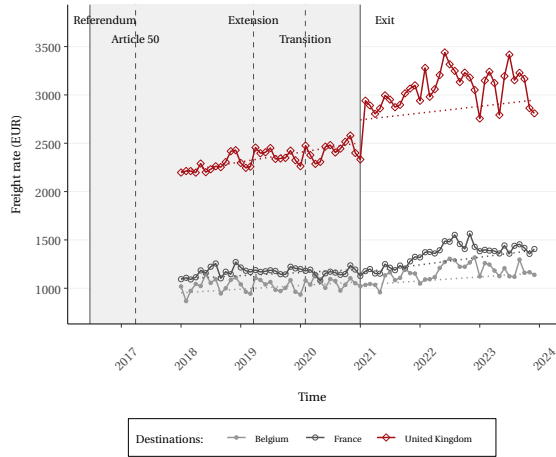
APPENDIX

The story of the paper in two graphs

(a) Mean consumer prices per destination



(b) Mean freight rate per destination



GBP/EUR Exchange Rate - [▶ Back](#)

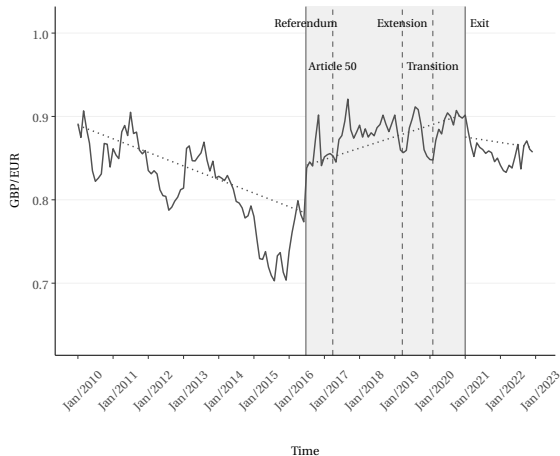


Figure 9: Source: European Central Bank

Daily Covid-19 infections per 1mln people - [▶ Back](#)

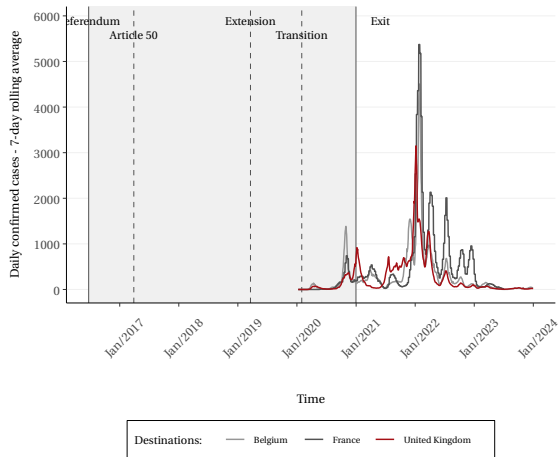


Figure 10: Source: Our world in Data - Oxford Martin School

Evolution of wage cost in retail - [▶ Back](#)

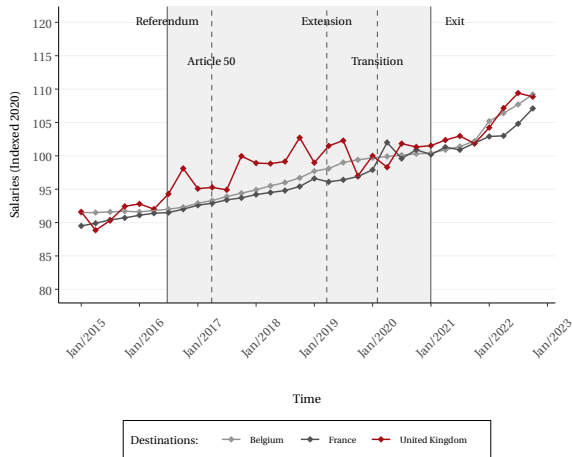


Figure 11: Source: Eurostat and Office for National Statistics (ONS)

Evolution of wage cost in retail - [▶ Back](#)

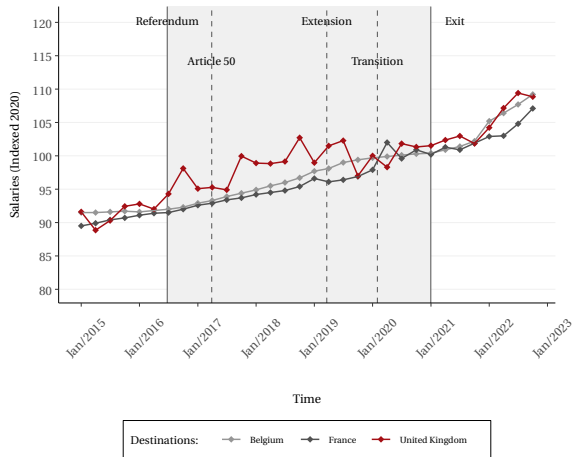


Figure 12: Source: Eurostat and Office for National Statistics (ONS)

Clarifying British Prices in the Wake of Brexit - domestically produced products - [▶ Back](#)

y	$\ln(p_{jt})$	$\ln(\hat{m}c_{jt})$	$\ln(\hat{\mu}_{jt})$	$\ln(mc_{jt}^{\text{Counter}})$	$\ln(p_{jt}^{\text{Counter}})$
	(1)	(2)	(3)	(4)	(5)
$T_{i,d} \times \mathbb{1}(t = 2018)$	0.0252** (0.00976)	0.0269*** (0.00981)	0.0167 (0.0108)	0.0429*** (0.0102)	0.0663*** (0.012)
$T_{i,d} \times \mathbb{1}(t = 2019)$	-	-	-	-	-
$T_{i,d} \times \mathbb{1}(t = 2020)$	-0.0182 (0.0116)	-0.0148 (0.0117)	-0.0389*** (0.012)	-0.0244** (0.0119)	-0.0134 (0.0134)
$T_{i,d} \times \mathbb{1}(t = 2021)$	0.0346 (0.0311)	0.0306 (0.0303)	0.0526 (0.0379)	0.00873 (0.0309)	0.0475 (0.033)
$T_{i,d} \times \mathbb{1}(t = 2022)$	0.151*** (0.0321)	0.141*** (0.0307)	0.2*** (0.0438)	0.121*** (0.0314)	0.144*** (0.0337)
Median of LHS	0.39	0.35	0.04	0.35	0.39
θ_i	✓	✓	✓	✓	✓
λ_t	✓	✓	✓	✓	✓
R^2	0.110	0.097	0.168	0.084	0.097
No. obs	1,493	1,493	1,493	1,493	1,493

Notes: Standard errors clustered at the product-destination level.
Significance at the 0.1*, 0.05**, 0.01*** levels.

▶ Same **specification** as before:

$$\ln(y_{j,t}) = \sum_t \beta_t (\mathbb{1}(d(j) = UK) \times \mathbb{1}(t \neq 2019M12)) + \theta_j + \lambda_t + \varepsilon_{j,t}$$

▶ **Actual** results:

$mc_{j,d,t}$ of domestic products 17% ↑

▶ **Counterfactual** results:

▶ $mc_{j,d,t}$ of domestic products 11.6% ↑

▶ $p_{j,d,t}$ of domestic products 12.3% ↑

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